

Disaster Recovery Plan

YAT Learning Management System (On-Premises)

System	YAT Learning Management System (on-premises)
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Prepared by	YAT ICT
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Contents

Contents	2
1. Executive Summary.....	3
2. Engagement Context and Scope.....	3
3. Current Recovery Position	3
3.1 Backup arrangements.....	3
3.2 Suppliers and dependencies	3
4. Impact Analysis.....	4
4.1 Recovery objectives	4
4.2 Data managed.....	4
4.3 Risk assessment.....	4
4.4 Exclusions	4
5. Recovery Strategy	4
5.1 Current strategy	4
5.2 Assessment against the objectives	4
5.3 Improvement direction	5
6. The Disaster Recovery Plan	5
6.1 Detection and declaration.....	5
6.2 Recovery steps.....	5
6.3 Meeting the recovery objectives	5
7. Plan Validation and Approval	5
7.1 Review	5
7.2 Sign-off	5
Document control	6

1. Executive Summary

This plan sets out the disaster recovery arrangements for the YAT on-premises Learning Management System (LMS). The LMS is backed up in full every night to tape, with tapes rotated offsite, and can be fully recovered from backup following a major incident such as server hardware failure, ransomware, or irrecoverable database corruption.

Current recovery performance — a worst-case Recovery Time Objective of 7–11 hours and a Recovery Point Objective of up to 24 hours — sits materially below the ICT Strategic Plan targets of $RTO \leq 4$ hours and $RPO \leq 1$ hour. The dominant gaps are the offsite tape-retrieval step (RTO) and the nightly-only backup cadence (RPO). Closing these gaps is an explicit driver of the planned LMS cloud migration.

2. Engagement Context and Scope

The YAT LMS (the DOODLE application on Windows Server 2016 with a MySQL database) runs on-premises at the Cremorne campus and is a mission-critical system for staff and students. This plan covers disaster recovery for that system — the backup arrangements, the recovery steps, and the recovery objectives currently achieved.

In scope: the LMS server, its MySQL database, the LMS application, and the LMS attachments (course materials and student submissions). Out of scope: unrelated campus systems, which are covered by their own arrangements, and any future cloud-hosted LMS, which is not yet in place.

3. Current Recovery Position

3.1 Backup arrangements

A complete, full backup of the LMS environment runs nightly (22:00 – 04:00 local time), captured to tape by the campus System Management server. The backup covers the Windows Server 2016 system state, the MySQL database (DOODLE schema and content), the LMS application binaries and configuration, and the LMS attachments. Tapes are rotated offsite each business day under the Backup and Retention Policy. The LMS remains available during the backup window; the database is captured against a brief consistency snapshot.

3.2 Suppliers and dependencies

Recovery depends on the offsite tape-storage provider (tape retrieval to the Cremorne campus takes 4–8 hours), on the availability of replacement server hardware, and on the campus Active Directory for post-restore authentication. There is no cloud or vendor-provided disaster-recovery capability for the on-premises LMS.

4. Impact Analysis

4.1 Recovery objectives

The ICT Strategic Plan sets the targets at $RTO \leq 4$ hours and $RPO \leq 1$ hour. The current on-premises process achieves a worst-case RTO of 7–11 hours and an RPO of up to 24 hours — both materially short of target.

Objective	Current achievement	Target
RTO — worst case	7–11 hours from incident declaration to LMS operational	≤ 4 hours
RPO — worst case	Up to 24 hours (anything written since the last nightly backup)	≤ 1 hour

4.2 Data managed

The protected data comprises the MySQL DOODLE database (schema and content) and the LMS attachments — course materials and student submissions. This includes student personal information and academic records, handled under the Privacy / Data Handling Policy and the Privacy Act.

4.3 Risk assessment

#	Risk event	Likelihood	Impact	Severity	What it drives
RE1	LMS server hardware failure	Medium	High	High	Restore to replacement hardware
RE2	Ransomware / malware compromise	Medium	Critical	High	Clean restore from offsite tape
RE3	Irrecoverable database corruption or deletion	Medium	High	High	Database restore from nightly backup
RE4	Campus-level event (fire / flood at Cremonne)	Low	Critical	High	Offsite tape copy; single-site exposure

4.4 Exclusions

This plan excludes unrelated campus systems (covered by their own arrangements) and routine single-file or partial restores (handled operationally). It assumes the offsite tapes are intact and retrievable, and that replacement hardware can be sourced.

5. Recovery Strategy

5.1 Current strategy

The current strategy is restore-from-tape onto replacement hardware at the Cremonne campus: a single-site, tape-based recovery. It relies entirely on the nightly tape backup and the offsite rotation for any protection against a campus-level event.

5.2 Assessment against the objectives

The strategy does not meet the targets. The offsite tape-retrieval step (4–8 hours) is the dominant contributor to the RTO gap; the nightly-only backup cadence is the dominant contributor to the RPO gap;

and a single campus site offers no rapid recovery option for a site-level disaster. The arrangement is adequate as a last-resort restore but cannot meet the strategic RTO/RPO targets.

5.3 Improvement direction

Closing the RTO/RPO gap is an explicit driver of the planned LMS cloud migration. A cloud-hosted platform is expected to provide faster recovery and a tighter recovery point than tape allows; the forward design work for the migrated LMS is required to demonstrate how the targets will be met. This plan will need to be replaced once the LMS is migrated.

6. The Disaster Recovery Plan

6.1 Detection and declaration

A major LMS incident is identified through system monitoring and the ICT helpdesk. The ICT Manager declares a disaster and authorises the recovery, notifying affected stakeholders.

6.2 Recovery steps

To completely recover the LMS from backup, the following steps are executed in sequence:

Step	Action	Owner	Duration
1	Request the relevant tape from offsite storage and have it delivered to the Cremorne campus	YAT ICT	4–8 hours
2	Restore the Windows Server 2016 system state and LMS application binaries from tape onto replacement hardware	YAT ICT	1 hour
3	Restore the MySQL database from the latest nightly backup and verify schema integrity	YAT ICT	1 hour
4	Verify LMS connectivity to MySQL, re-authenticate against Active Directory, and confirm end-user sign-in	YAT ICT	1 hour

6.3 Meeting the recovery objectives

The plan does not currently meet the strategic objectives: the cumulative recovery time of 7–11 hours exceeds the $RTO \leq 4$ hours, and the nightly backup cadence allows up to 24 hours of data loss against the $RPO \leq 1$ hour. The gap is documented here as the basis for the improvement work.

7. Plan Validation and Approval

7.1 Review

This plan is reviewed annually, or on a material change to the LMS environment or backup tooling, and is walked through with YAT ICT as part of the review.

7.2 Sign-off

Role	Name	Date	Acceptance
Prepared by	YAT ICT	[date]	Issued
Approved by	Sam Walker (YAT ICT Manager)	[date]	Current DR arrangements accepted; gap to target noted

Document control

Field	Value
Document version	v2.3
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Review cycle	Annual, or on material change to the LMS environment or backup tooling
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Related documents	Backup and Retention Policy; ICT Strategic Plan; LMS Server Specifications and Current Status